Loan Risk Prediction

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Problem Statement

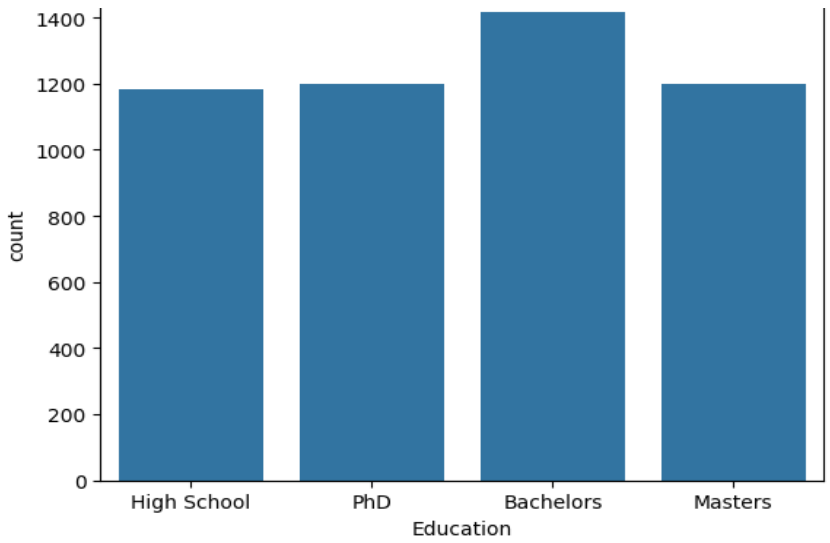
The objective of this project is to develop a machine learning model that can accurately predict whether a loan applicant is likely to be approved or rejected based on their financial and personal attributes. This will help in making faster, data-driven, and reliable loan approval decisions while minimizing risk

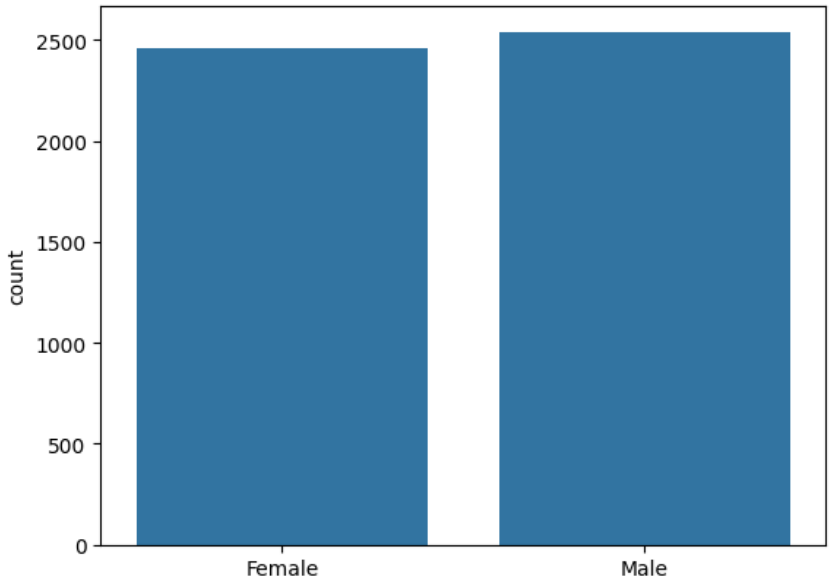
Dataset Overview

5000 rows and 10 columns are in the dataset.

Handled missing values using mean for numeric values and mode for categorical values

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Age              5000 non-null   int64
 1   Income           4804 non-null   float64
 2   LoanAmount       5000 non-null   float64
 3   CreditScore      4806 non-null   float64
 4   YearsExperience  5000 non-null   int64
 5   Gender           5000 non-null   object
 6   Education        4802 non-null   object
 7   City             5000 non-null   object
 8   EmploymentType   5000 non-null   object
 9   LoanApproved     5000 non-null   int64
dtypes: float64(3), int64(3), object(4)
memory usage: 390.8+ KB
```





Exploratory Data Analysis (EDA)

- Male and female are almost equally applying for the loan
- Most of the of loan applicants have a Bachelor’s degree

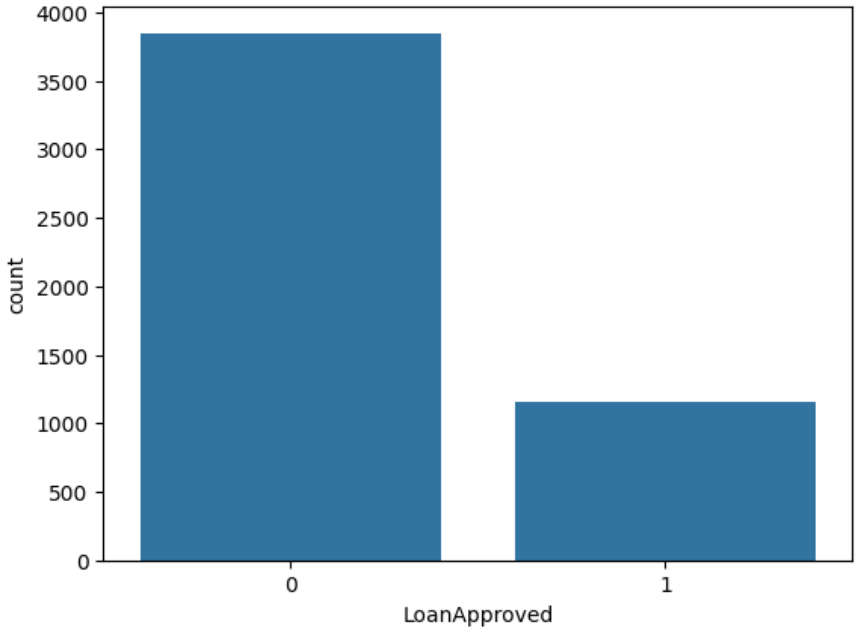
Target Variable Distribution

- The dataset is highly imbalanced, with significantly more rejected applications than approved ones

- Since the dataset was imbalanced, the model could become biased toward predicting rejections. To solve this, I used SMOTE, which creates synthetic data points for the minority class, helping the model learn both classes equally

Before

LoanApproved	
0	3849
1	1151



After

LoanApproved	
0	3849
1	3849

Number of outlier:

	columns	No of outliers
0	Age	0
1	Income	47
2	LoanAmount	32
3	CreditScore	0
4	YearsExperience	0
5	LoanApproved	1151

I found outliers in income and loan amount, but I did remove it because in financial data extreme values a realistic

Checking this heatmap can clearly see year of experience and city has almost zero correlation with every features so, I dropped before modeling

Model Building

Trained multiple models

- **Logistic regression**
- **KNN classifier**
- **Naïve bias classifier**
- **Decision tree classifier**
- **Random forest classifier**
- **Ada boost classifier**
- **Gradient boosting classifier**
- **Bagging classifier**
- **XGB classifier**

Logistic regression

- Overall accuracy is only 79%
- Class 1 performance is too low

```
Classification report:

Train Score:
0.8034285714285714
              precision    recall  f1-score   support

           0       0.83      0.90      0.87      1142
           1       0.58      0.43      0.49       358

    accuracy                           0.79      1500
   macro avg       0.71      0.67      0.68      1500
weighted avg       0.77      0.79      0.78      1500
```

Decision Tree Classifier

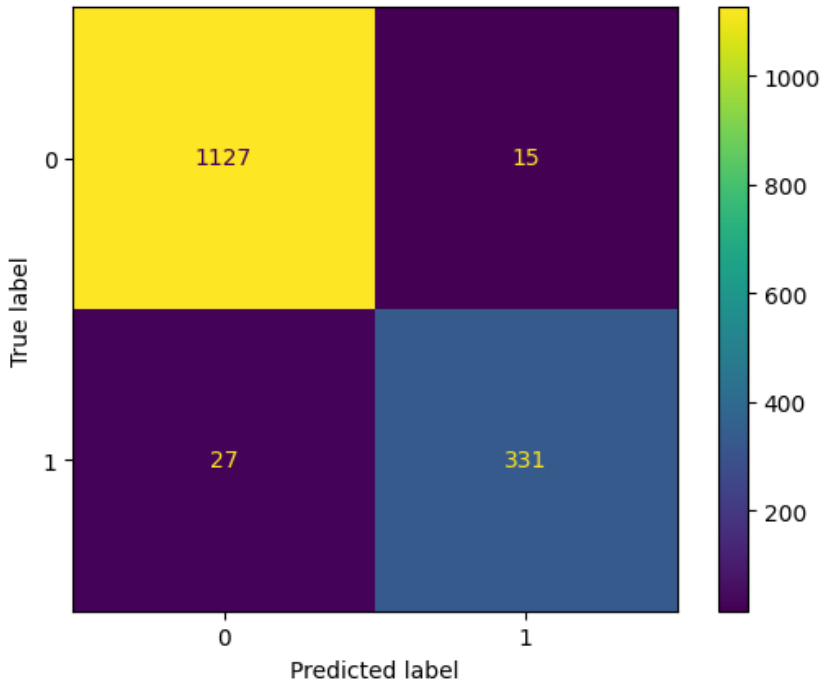
- Test accuracy is 94% and train score is 1
- Model is overfitting

```
Train Score:
1.0
              precision    recall  f1-score   support

           0       0.97      0.95      0.96      1142
           1       0.85      0.90      0.88       358

    accuracy                           0.94      1500
   macro avg       0.91      0.93      0.92      1500
weighted avg       0.94      0.94      0.94      1500
```

Best Model Selection

Among all the models tested, the Gradient Boosting Classifier achieved the best performance.

- Pervious model had train score of 99% and test 96% slightly overfitted
- But Gradient Boosting is perfect balanced in test and train with 97%
- High precision and recall moreover, balanced performance in both classes
- All ensemble models demonstrated superior performance compared to individual models




```
Classification report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.98      1142
           1       0.96      0.92      0.94       358

    accuracy                           0.97      1500
   macro avg       0.97      0.96      0.96      1500
weighted avg       0.97      0.97      0.97      1500
```

Deployment

- The trained Gradient Boosting model was serialized using Pickle and integrated into the Streamlit application for real-time predictions

- I deployed the model using Streamlit, where users can input their details and get real-time loan approval predictions. The model is loaded using Pickle and processes inputs before generating results

Thank You